

query q $[H, d]$

router: $\rho_b = \max(q \cdot a_k, \max_j q \cdot R_{K,j}) \cdot \text{scale} \rightarrow \text{keep top-}K = 16 \text{ blocks}$

exact $q \cdot k$ over anchor
+ residual keys

Low-rank branch (selected blocks)

anchor score $s_{anc} = (q \cdot a_k) \cdot \text{scale}$

project query $\tilde{q} = (q V_K^\top) \cdot \text{scale} \in \mathbb{R}^{32}$

delta scores $\delta s = s(\tilde{q} U^\top) + s_{anc}$

per-block softmax $\rightarrow w, \text{lse}_{sp}$

$O_{sp} = \sum w a_v + (\sum w U) s V_V$

$O(KrB)$ — never forms $\hat{K}$

Exact branch (residuals + recency)

concat selected blocks' exact residuals
 $R_K, R_V \oplus$ dense recency window

**exact attention over the
augmented $\tilde{K}, \tilde{V}$ (full fp16)**

softmax $\rightarrow O_{dn}, \text{lse}_{dn}$

verbatim recall of
needle tokens

flash-style LSE merge

$m = \max(\text{lse}_{sp}, \text{lse}_{dn}); \quad o = (e^{\cdot O_{sp}} + e^{\cdot O_{dn}}) / \Sigma$

NaN/inf-guarded: an empty compressed set contributes exactly zero weight